

Ethics of Collecting User Data in Modern Web Applications

A Structured Literature Review and Synthesis of Nine Research Papers

Prepared as a Research Reference Document

Covers: Dark Patterns, Consent Design, Informed Consent, and User Autonomy

Part 1 — Individual Paper Summaries

Paper 1: Dark Patterns Post-GDPR: Scraping Consent Interface Designs and Demonstrating their Influence

Authors: Midas Nouwens, Ilaria Liccardi, Michael Veale, David Karger, Lalana Kagal

Methodology

A two-part study consisting of (1) web scraping and legal evaluation of the five most prevalent third-party Consent Management Platforms (CMPs) across the top 10,000 UK websites (n=680), and (2) a field experiment with 40 participants using a custom browser plugin to test how eight distinct consent interface designs affect actual user choices.

Key Findings

Deceptive user interfaces and implied consent options are universally common on the modern web. Only 11.7% of the scraped CMP interfaces met the absolute minimum legal requirements mandated under European data protection laws. The field experiment showed that minor design modifications drastically alter consent metrics — removing an explicit opt-out button boosts consent by 22–23 percentage points, while granular control checkboxes on the initial screen drop consent rates by 8–20 percentage points.

Direct Relevance to the Topic

Highlights the ethical breakdown of outsourcing data collection compliance to third-party software vendors. Provides empirical verification that digital developers actively use asymmetric interface layouts to undermine legal provisions (like the GDPR) and subvert user autonomy to maximize corporate data accumulation.

Paper 2: Dark Patterns at Scale: Findings from a Crawl of 11K Shopping Websites

Authors: Arunesh Mathur, Gunes Acar, Michael J. Friedman, Elena Lucherini, Jonathan Mayer, Marshini Chetty, Arvind Narayanan

Methodology

Automated text-clustering and crawling tools built on the OpenWPM privacy framework, simulating a real human purchasing journey across approximately 53,000 product pages on 11,000 e-commerce and shopping platforms.

Key Findings

Uncovered 1,818 instances of dark patterns spanning 15 unique design types and 7 structural categories, affecting roughly 11.1% of sampled sites. Higher-traffic websites (by Alexa ranking) were more likely to deploy manipulative elements. Identified 22 distinct third-party entities that sell dark patterns directly to platforms as premade, turnkey solutions.

Direct Relevance to the Topic

Exposes the systemic, industrialized marketplace underlying dark patterns. Proves user manipulation is not just isolated design choices, but a scalable commercial service designed to harvest personal information and capitalize on cognitive vulnerabilities.

Paper 3: The OK Is Not Enough: A Large Scale Study of Consent Dialogs in Smartphone Applications

Authors: Simon Koch, Benjamin Altpeter, Martin Johns

Methodology

An OS-agnostic testing pipeline applied to 3,006 Android and 1,773 iOS apps, examining the structure of privacy dialogs and measuring dynamic traffic to trackers before and after user interaction with consent buttons.

Key Findings

Only 22.3% of sampled mobile apps display any privacy notification upon launch; of those, only 11.9% give users an actionable choice. Most prompts use coercive styling. Tracking libraries regularly transmit data to third parties before any user interaction, or continue processing data despite explicit rejection.

Direct Relevance to the Topic

Expands the ethical scope from desktop into mobile ecosystems, documenting severe non-compliance where platforms prioritize tracking mechanisms above user consent — making the interface choice largely illusory.

Paper 4: Permission Rationales in the Web Ecosystem: An Exploration of Rationale Text and Design Patterns

Authors: Yusra Elbitar, Soheil Khodayari, Marian Harbach, Gianluca De Stefano, Balazs Csaba Engedy, Giancarlo Pellegrino, Sven Bugiel

Methodology

Large-scale analysis of 770,000+ webpages using an interactive Chromium-based crawler, leveraging LLMs and BERT classifiers to extract and categorize 3,674 permission rationale texts and 749 rationale UIs across 85,000 webpages, cross-referenced with Chrome telemetry data.

Key Findings

Web developers can trigger hardware prompts (notifications, camera, location) at any time without oversight. Contextual justification increases approval rates; positive tones boost approvals by 18%. Full-screen justifications increase approvals by 33%, and interface overlays by 41% — but these recurring modals also increase user annoyance and cognitive fatigue.

Direct Relevance to the Topic

Sheds light on the ethics of choice architecture, detailing how psychological framing, timed prompts, and modal designs are used to bypass natural privacy boundaries and capture sensitive hardware capabilities.

Paper 5: Circumvention by Design: Dark Patterns in Cookie Consents for Online News Outlets

Authors: Than Htut Soe, Oda Elise Nordberg, Frode Guribye, Marija Slavkovik

Methodology

A manual, granular examination of 300 data-gathering consent notices across Scandinavian and international English news applications, focused on interface features that automated crawlers often miss.

Key Findings

Every single surveyed news interface deployed some degree of unethical design. Introduced eight new contextual dark pattern variants, showing platforms deliberately increase interaction complexity (e.g. more clicks to reject than accept, confusing language) to dissuade opt-out.

Direct Relevance to the Topic

Highly relevant because it evaluates a sector — news media — that relies on societal trust rather than primarily trading in user data. That 100% of portals used manipulative configurations shows ad-revenue dependency compromises ethical values across the board.

Paper 6: Online Privacy and Informed Consent: The Dilemma of Information Asymmetry

Authors: Masooda Bashir, Carol Hayes, April D. Lambert, Jay P. Kesan

Methodology

A two-part survey at a large US academic institution: a quantitative knowledge evaluation (n=455) measuring technical comprehension, and an empirical behavior section (n=756) assessing privacy beliefs and attitudes.

Key Findings

Reveals severe information asymmetry between consumers and internet firms. Despite high education levels, major knowledge gaps exist regarding how data collection, cloud servers, and behavioral trading brokers process personal files. This, combined with manipulative "take-it-or-leave-it" contracts, drives the well-known privacy paradox.

Direct Relevance to the Topic

Introduces the core philosophical framework for the topic: true "informed consent" cannot exist under current conditions, since platforms exploit informational gaps via legalese that hinders comprehension and undermines voluntariness.

Paper 7: "I am Definitely Manipulated, Even When I am Aware of it. It's Ridiculous!" — Dark Patterns from the End-User Perspective

Authors: Kerstin Bongard-Blanchy, Arianna Rossi, Salvador Rivas, Sophie Doublet, Vincent Koenig, Gabriele Lenzini

Methodology

An end-user perception survey of 406 demographically representative UK individuals, assessing baseline awareness, usage patterns, and the ability to spot manipulative designs in ten unbranded interface mockups under timed conditions.

Key Findings

Users are generally aware manipulative designs exist and intend to influence behavior, but this awareness does not translate into practical resistance. Younger users especially can recognize a design as "unfair," yet remain uncertain about the actual downstream harm, leaving them vulnerable to short-term gratification

over long-term privacy protection.

Direct Relevance to the Topic

Highlights that user awareness alone is an insufficient defense. Interfaces exploit cognitive biases effectively enough that even recognizing manipulation doesn't stop users from yielding to it, due to friction, learned helplessness, and immediate gratification.

Paper 8: Shadows in the Interface: A Comprehensive Study on Dark Patterns

Authors: Liming Nie, Yangyang Zhao, Chenglin Li, Xuqiong Luo, Yang Liu

Methodology

Development of the Dark Pattern Analysis Framework (DPAF): a systematic review of 76 academic papers to build an industry-validated taxonomy of 64 dark pattern types, then assessing detection tools and datasets against this taxonomy.

Key Findings

Prior classifications are disjointed and incomplete. Of five modern detection tools tested, only 32 of 64 pattern categories were successfully identified — a 50% coverage gap. The four largest public datasets used for training ML/computer-vision models also completely omitted 32 of the 64 types.

Direct Relevance to the Topic

Reveals the technical limitations of protecting users from automated, predatory data harvesting, showing current algorithmic systems are insufficient and highlighting the need for structural design principles and policy reform.

Paper 9: Influencing User Decisions: Dilemmas in Designing Online Interfaces

Authors: Marie-Sophie Simon, Hanna Schraffenberger, Raphaël Gellert

Methodology

A philosophical review and theoretical analysis of the distinction between acceptable choice architectures (nudges) and unacceptable, predatory designs (manipulation/dark patterns), mapping three practical design dilemmas faced by well-meaning developers.

Key Findings

The conceptual boundary between a "nudge" and a "dark pattern" is highly inconsistent and contradictory across the literature. Both frameworks assume users are irrational actors plagued by cognitive biases — a nudge corrects this bias for the user's benefit, a dark pattern exploits it for profit. The authors argue this shared worldview diminishes user agency.

Direct Relevance to the Topic

Addresses the core design philosophy of the topic: ethical data collection requires moving away from paternalistic choice architecture that exploits subconscious flaws, toward design that treats users as rational, independent agents.

Part 2 — Holistic Synthesis and Interconnection

Viewed together as a single body of work, these nine papers map out the technical, psychological, legal, and systemic dimensions of the ethics of collecting user data in modern web applications. Below is how the concepts across all nine documents connect to tell a unified story.

1. The Core Ethical Issue: Information Asymmetry and the Illusion of Choice

The foundation of web-based data collection relies on the ethical ideal of "informed consent." However, research shows this process is systematically compromised. Paper 6 outlines the root problem: an overwhelming information asymmetry where users lack technical understanding of where their data goes, while platforms hide behind complex legal terminology — turning consent into an all-or-nothing "take-it-or-leave-it" requirement. Paper 9 adds a philosophical point: even well-meaning developers often treat users as irrational actors, creating interfaces that manage choices rather than respect independent human agency.

2. The Weaponization of User Interfaces (Dark Patterns)

Because web applications face intense pressure to capture consumer data for monetization, they systematically use interface layout to exploit human psychological vulnerabilities. Paper 8's taxonomy of 64 distinct dark pattern types formalizes this. Papers 1 and 5 demonstrate how this functions in daily browsing — websites use visual asymmetry, hidden opt-out toggles, and complex interaction paths to make declining data collection as difficult as possible. Paper 4 details how applications use intrusive overlays and strategic full-screen rationales to push users into granting expansive hardware permissions (location, camera, notifications).

3. The Industrialization of Data Extraction

Predatory user experiences are rarely accidental; they are industrialized corporate strategies. Paper 2 identifies third-party companies that sell turnkey dark pattern modules directly to online businesses, proving manipulation is a scalable commercial service. This directly affects compliance outsourcing — as shown in Paper 1, even when websites integrate dedicated Consent Management Platforms (CMPs) to fulfill laws like the GDPR, these platforms are systematically configured to prioritize data harvesting over genuine legal compliance.

4. The Breakdown of Defenses and Technical Exploitation

The ultimate ethical breakdown occurs when user controls are bypassed entirely. Paper 7 shows that even highly aware users who want to protect their privacy still surrender their data due to interface friction and learned fatigue. This is compounded by technical misconduct: Paper 3's app telemetry shows tracking scripts frequently leak sensitive data to third parties before any user interaction, or ignore explicit opt-out decisions entirely. Finally, Paper 8 warns that automated auditing is falling behind — modern detection tools and ML models capture only about 50% of manipulative behaviors, leaving a significant gap in our ability to monitor and enforce data ethics.

Summary Conclusion

Taken together, this body of work shows that the ethics of collecting user data cannot be solved by simply requiring a platform to display a cookie banner or an "agree" checkbox. True data ethics requires a fundamental shift in design philosophy: web applications must move away from predatory, asymmetric interfaces that exploit human cognitive limitations for financial gain, and instead implement transparent design structures that reduce information asymmetry, respect user autonomy, and support independent, deliberate decision-making.